

Md. Mehedi Hasan

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Academic Credentials

Khulna University of Engineering & Technology (KUET) | Khulna-6600, Bangladesh [July 2024-present]

Master of Science (M.Sc. Eng.) in Electrical and Electronic Engineering

CGPA: **3.83 / 4.00** | **Coursework:** Completed | **Thesis Proposal:** Approved | **Expected Completion:** 2026

Thesis Title: *A Multi-Level Control Framework for Voltage Quality Improvement in PV-Integrated Distribution Networks*

Supervisor: Naruttam Kumar Roy, Ph.D. [Google Scholar](#)

Pabna University of Science and Technology (PUST) | Pabna-6600, Bangladesh [March 2018-April 2024]

Bachelor of Science (B.Sc) in Electrical and Electronic Engineering

CGPA: **3.85 / 4.00** (Ranked **1st** in class of 43 students)

Thesis Title: *Management of EVs for Mitigating Unbalance in a LV Distribution Grid*

Supervisor: Md. Rabiul Islam, Ph.D. [Google Scholar](#)

Teaching Experience

Khulna Khan Bahadur Ahsanullah University | Khulna-6600, Bangladesh [1 september 2024-Present]

Lecturer – Department of Electrical and Electronic Engineering

[Faculty Profile](#)

Research Interest

- Power System Stability.
- Smart Grids.
- Renewable Energy Integration.
- Electric Vehicle.
- Microgrid.

Journal Publications

- **M. M. Hasan**, K.A. Refat, M.F. Ali, M.S. Alam, O. Obaidi, M. Ali, M.K. Islam, I.H. Imran (2026). "Design and techno-economic validation of a university campus hybrid microgrid using HOMER Pro and DIgSILENT PowerFactory". Scientific Reports, 16, 17248 (2026). <https://doi.org/10.1038/s41598-026-45872-9> [Link](#)
- M. G. Hasan, M. F. Rahman, **M. M. Hasan**, M. Hasan, M. A. Uddin, M. G. Sadeque, A. H. M. I. Ferdous, Sustainable charging framework: Solar wireless EV charging enhanced with MPPT control, Energy Reports, Volume 15, 2026, 109274, ISSN 2352-4847, <https://doi.org/10.1016/j.egy.2026.109274>. [Link](#)

Under Review / Submitted

- S. Das, **M. M. Hasan***, M. S. Alam, A. H. Taj "A Unified Framework for Hybrid Renewable Energy Microgrid Planning and Operational Feasibility Assessment," International Journal of Energy Research, Wiley (**Under Review**).
- **M. M. Hasan**, A. H. Shakil, M. R. Islam, and N. K. Roy, "A Hierarchical Control Framework for Voltage Unbalance and Neutral Current Mitigation in PV-Rich Low-Voltage Distribution Networks". (**M.Sc. thesis**), International Journal of Energy Research, Wiley (**Under Review**).
- M. M. Hasan, A. H. Shakil, N. K. Roy, and M. R. Islam, "Optimal Phase Placement and PV Curtailment for Voltage Unbalance Mitigation in PV-Rich Low-Voltage Networks: A Comparative Study of GA, PSO, and DE". (**M.Sc. thesis**), Renewable Energy, Elsevier (**Submitted**).

Conferences

- **M. M. Hasan**, M. A. Yousuf and M. R. Islam, "Management of EVs for Mitigating Unbalance in a LV Distribution Grid," 2024 6th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT), Dhaka, Bangladesh, 2024, pp. 1152-1156, doi: 10.1109/ICEEICT62016.2024.10534474. [Link](#)
- M. R. Rashid, **M. M. Hasan*** and M. R. Islam, "Develop a Method for Mitigating Unbalance Considering EV Users Transportation Needs," 2025 International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN), Rangpur, Bangladesh, 2025, pp. 1-5, doi: 10.1109/QPAIN66474.2025.11171788. [Link](#)
- **M. M. Hasan**, M. R. Islam and M. A. Yousuf, "Mitigating Unbalance Due to EV Dispatch in a LV Distribution Grid Having Balanced Loads," 2024 IEEE International Conference on Power, Electrical, Electronics and Industrial Applications (PEEIACON), Rajshahi, Bangladesh, 2024, pp. 1-4, doi: 10.1109/PEEIACON63629.2024.10800142. [Link](#)
- **M. M. Hasan***, A. H. Shakil, M. R. Islam, and N. K. Roy, "Coordinated Phase Rebalancing of Rooftop PV Systems for Voltage Imbalance Mitigation in LV Networks," International Conference on Power, Electronics, Communications, Computing, and Intelligent Infrastructure 2026 (PECCII), PUST, Pabna, Bangladesh, Jun. 17-18, 2026 (**Presented**).

Supervised

- M. W. Ali, **M. M. Hasan***, M. A. Yousuf, and M. I. Ahmad, "Optimal Capacitor Placement and Sizing in Unbalanced Distribution Systems Using a Chaotic Iterative Local Search with Adaptive Descent Algorithm, International Conference on Power, Electronics, Communications, Computing, and Intelligent Infrastructure 2026 (PECCII), PUST, Pabna, Bangladesh, Jun. 17-18, 2026 (**Presented**).
- R. Islam, M. A. M. Jesun, M. A. Yousuf, **M. M. Hasan***, and M. S. Alam, "Optimization and Feasibility Analysis of a Grid-Connected Hybrid Renewable Microgrid for Cost and Emission Reduction in Bangladesh," International Conference on Power, Electronics, Communications, Computing, and Intelligent Infrastructure 2026 (PECCII), PUST, Pabna, Bangladesh, Jun. 17-18, 2026 (**Presented**).
- P. Chowdhury, S. Das, M. A. Yousuf, **M. M. Hasan***, and Md. F. Ali, "Techno-Economic Optimization of a Grid-Connected PV-Wind-BESS Microgrid for Commercial Loads in Bangladesh," International Conference on Power, Electronics, Communications, Computing, and Intelligent Infrastructure 2026 (PECCII), PUST, Pabna, Bangladesh, Jun. 17-18, 2026 (**Presented**).

Skills & Expertise

- **Programming Language:** C, C++, DIgSILENT Programming Language (DPL) — scripting and automation.
- **Softwares:** DIgSILENT PowerFactory, PSSE, MATLAB, Edraw Max, PVsyst, CircuitLab, Simulink, Multisim, 4nec2.
- **Productivity & Office Tools:** Microsoft Excel, PowerPoint, Word, Visio.

Projects

- Developed a detailed model of the Southern Bangladesh power network based on the PGCB network using PSS®E and performed power-flow analysis to evaluate voltage profiles, frequency, active/reactive power flows, and overall network performance.
- Design a system prototype using arduino for preventing train accidents at rail crossing.
- Earthquake Monitor and Alerting System.

Supervised

- An IoT-Based Intelligent Traffic Management and Control System.
- Automatic Water Level Control System for Water Tanks.
- Password-Authenticated Digital Security Lock System using an Arduino Uno.

Internship

- Intern, "Rajbari Palli Bidyut Samity, Rajbari, Bangladesh". [April 2023]
- Industrial visit at "Bangladesh India Power Transmission Centre, HVDC, Bheramara, Kushtia". [August 2022]
- Industrial visit at "Khulna 225MW Combined Cycle Power Plant, Khulna". [July 2026]

Awards & Certificates

- **Ranked 1st in the Engineering Faculty** in the B.Sc. Examination Results at PUST.
- **Dean's List Award** for academic excellence (CGPA: 3.85, Session: 2017–18).
- **Student of the Year Award** for three consecutive years (1st, 2nd, and 3rd Year B.Sc. Examinations).
- **University Merit Scholarship**, Pabna University of Science and Technology (PUST).
- Served as **Class Representative for four consecutive years**, contributing to effective communication, coordination, and student leadership.

References

Dr. Md. Rabiul Islam

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Dr. Naruttam Kumar Roy

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